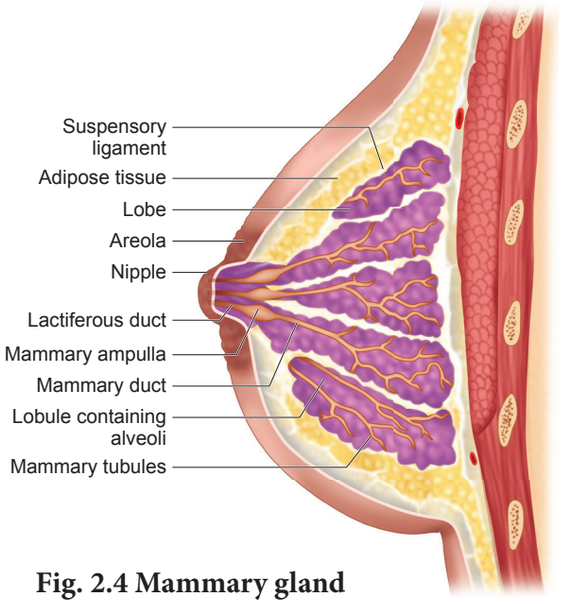


Fig. 2.4 Mammary gland

up of lobules which contain **acini or alveoli** lined by epithelial cells. Cells of the alveoli secrete milk. The alveoli open into **mammary tubules**. The tubules of each lobe join to form a **mammary duct**. Several mammary ducts join to form a wider **mammary ampulla** which is connected to the **lactiferous duct** in the nipple. Under the nipple, each lactiferous duct expands to form the **lactiferous sinus** which serves as a reservoir of milk. Each lactiferous duct opens separately by a minute pore on the surface of the nipple.

Normal development of the breast begins at puberty and progresses with changes during each menstrual cycle. In non-pregnant women, the glandular structure is largely underdeveloped and the breast size is largely due to amount of fat deposits. The size of the breast does not have an influence on the efficiency of lactation.

2.2 Gametogenesis

Gametogenesis is the process of formation of gametes i.e., sperms and ovum from the primary sex organs in all sexually reproducing organisms. Meiosis plays the most significant role in the process of gametogenesis **(Fig.2.5)**.

Spermatogenesis

Spermatogenesis is the sequence of events in the seminiferous tubules of the testes that produce the male gametes, the sperms. During development, the primordial germ cells migrate into the testes and become immature germ cells called sperm mother cells or spermatogonia in the inner surfaces of the seminiferous tubules **(Fig. 2.6 a)**. The spermatogonia begin to undergo mitotic division at puberty and continue throughout life.

In the first stage of spermatogenesis, the spermatogonia migrate among sertoli cells towards the central lumen of the seminiferous tubule and become modified and enlarged to form primary spermatocytes which are diploid with 23 pairs i.e., 46 chromosomes.

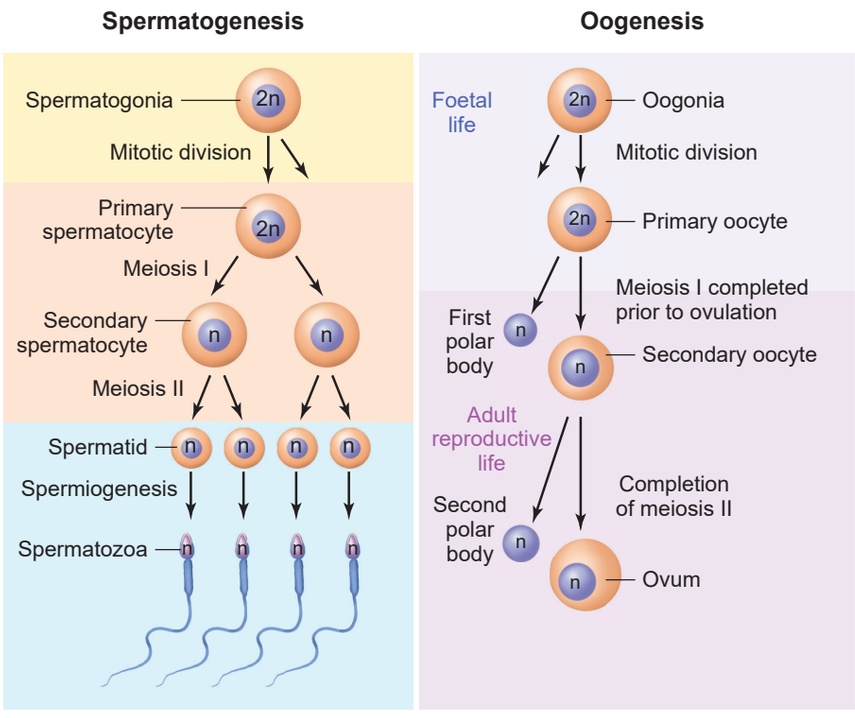


Fig. 2.5 Gametogenesis

XII Std Zoology Chapter 2 EM.indd 16 23/12/2022 14:52:30